

In [1]: `pip install seaborn`

Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: seaborn in c:\programdata\anaconda3\lib\site-packages (0.13.2)
Requirement already satisfied: numpy!=1.24.0,>=1.20 in c:\programdata\anaconda3\lib\site-packages (from seaborn) (2.3.5)
Requirement already satisfied: pandas>=1.2 in c:\programdata\anaconda3\lib\site-packages (from seaborn) (2.3.3)
Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in c:\programdata\anaconda3\lib\site-packages (from seaborn) (3.10.6)
Requirement already satisfied: contourpy>=1.0.1 in c:\programdata\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.3.3)
Requirement already satisfied: cycler>=0.10 in c:\programdata\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\programdata\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\programdata\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.4.9)
Requirement already satisfied: packaging>=20.0 in c:\programdata\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (25.0)
Requirement already satisfied: pillow>=8 in c:\programdata\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (12.0.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\programdata\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (3.2.5)
Requirement already satisfied: python-dateutil>=2.7 in c:\programdata\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\programdata\anaconda3\lib\site-packages (from pandas>=1.2->seaborn) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in c:\programdata\anaconda3\lib\site-packages (from pandas>=1.2->seaborn) (2025.2)
Requirement already satisfied: six>=1.5 in c:\programdata\anaconda3\lib\site-packages (from python-dateutil>=2.7->matplotlib!=3.6.1,>=3.4->seaborn) (1.17.0)
Note: you may need to restart the kernel to use updated packages.

In [2]: `import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns`

In [3]: `titanic = sns.load_dataset('titanic')`

In [4]: `titanic.head()`

Out[4]:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN



In [5]: `titanic.tail()`

Out[5]:

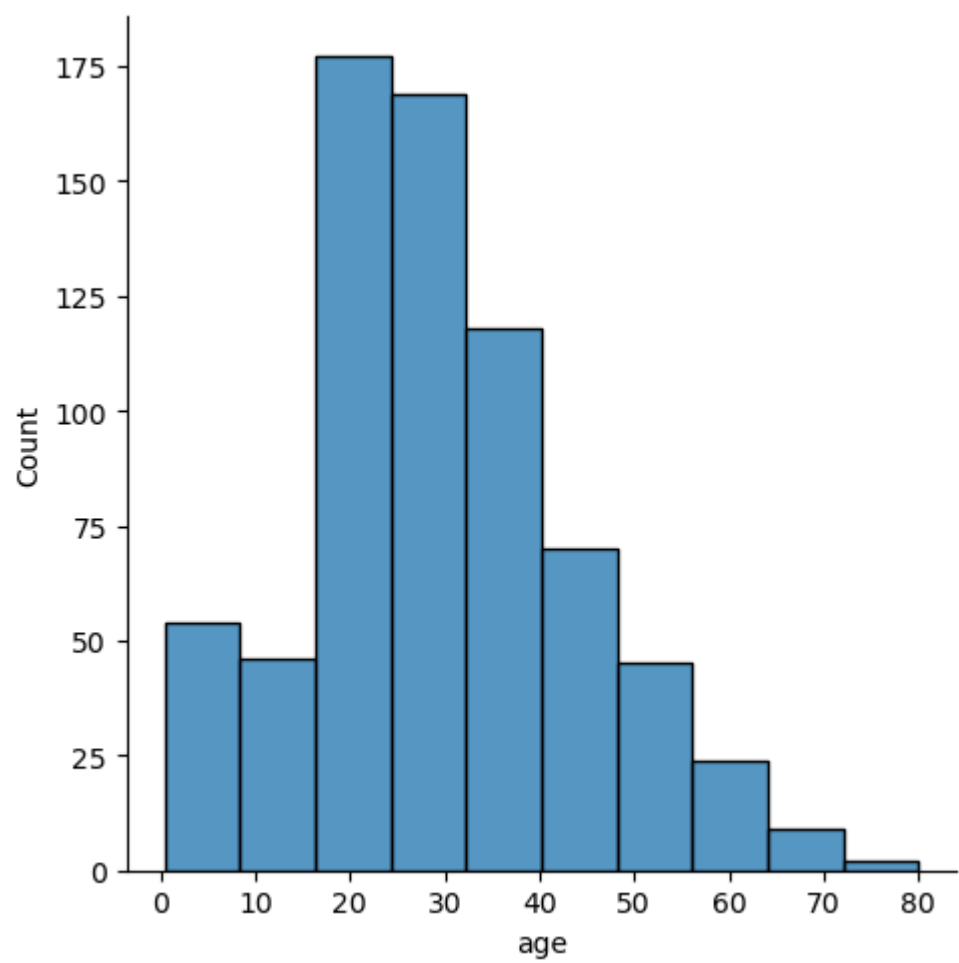
	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck
886	0	2	male	27.0	0	0	13.00	S	Second	man	True	NaN
887	1	1	female	19.0	0	0	30.00	S	First	woman	False	E
888	0	3	female	NaN	1	2	23.45	S	Third	woman	False	NaN
889	1	1	male	26.0	0	0	30.00	C	First	man	True	C
890	0	3	male	32.0	0	0	7.75	Q	Third	man	True	NaN

In [6]:

```
sns.displot(titanic['age'], bins = 10)
```

Out[6]:

<seaborn.axisgrid.FacetGrid at 0x26a2e4bfa10>

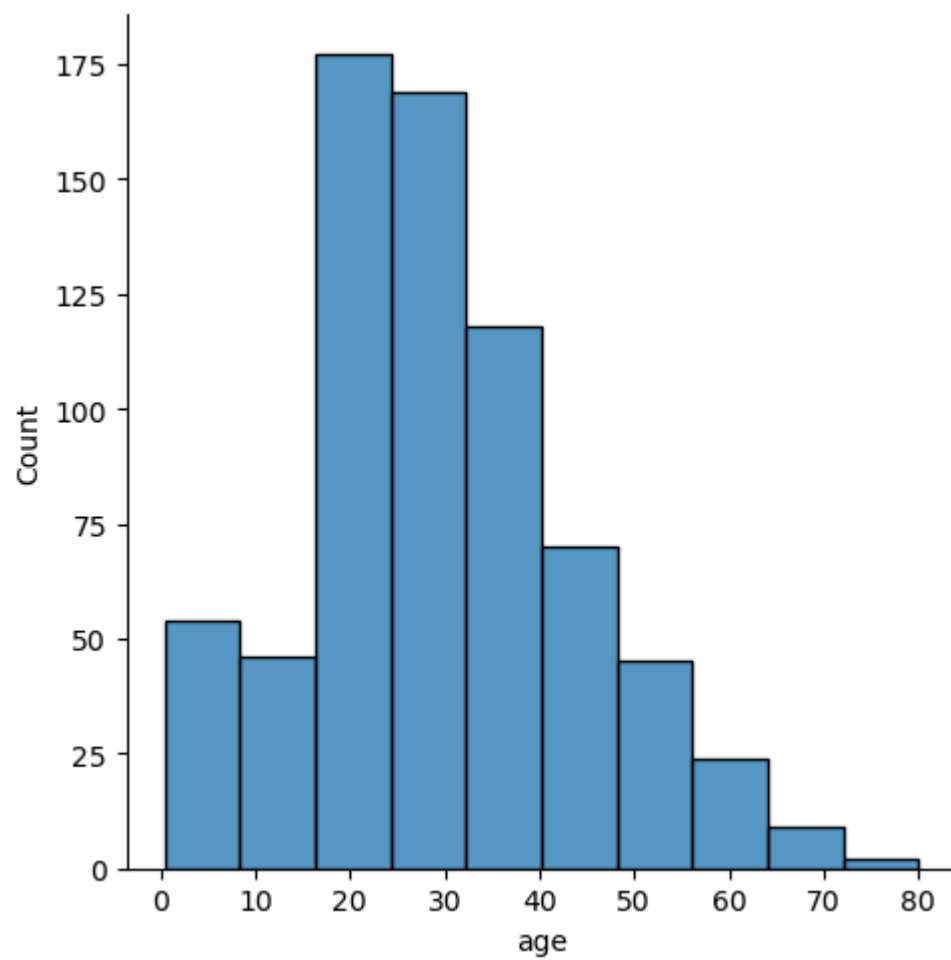


In [7]:

```
sns.displot(titanic['age'], bins = 10, kde = False)
```

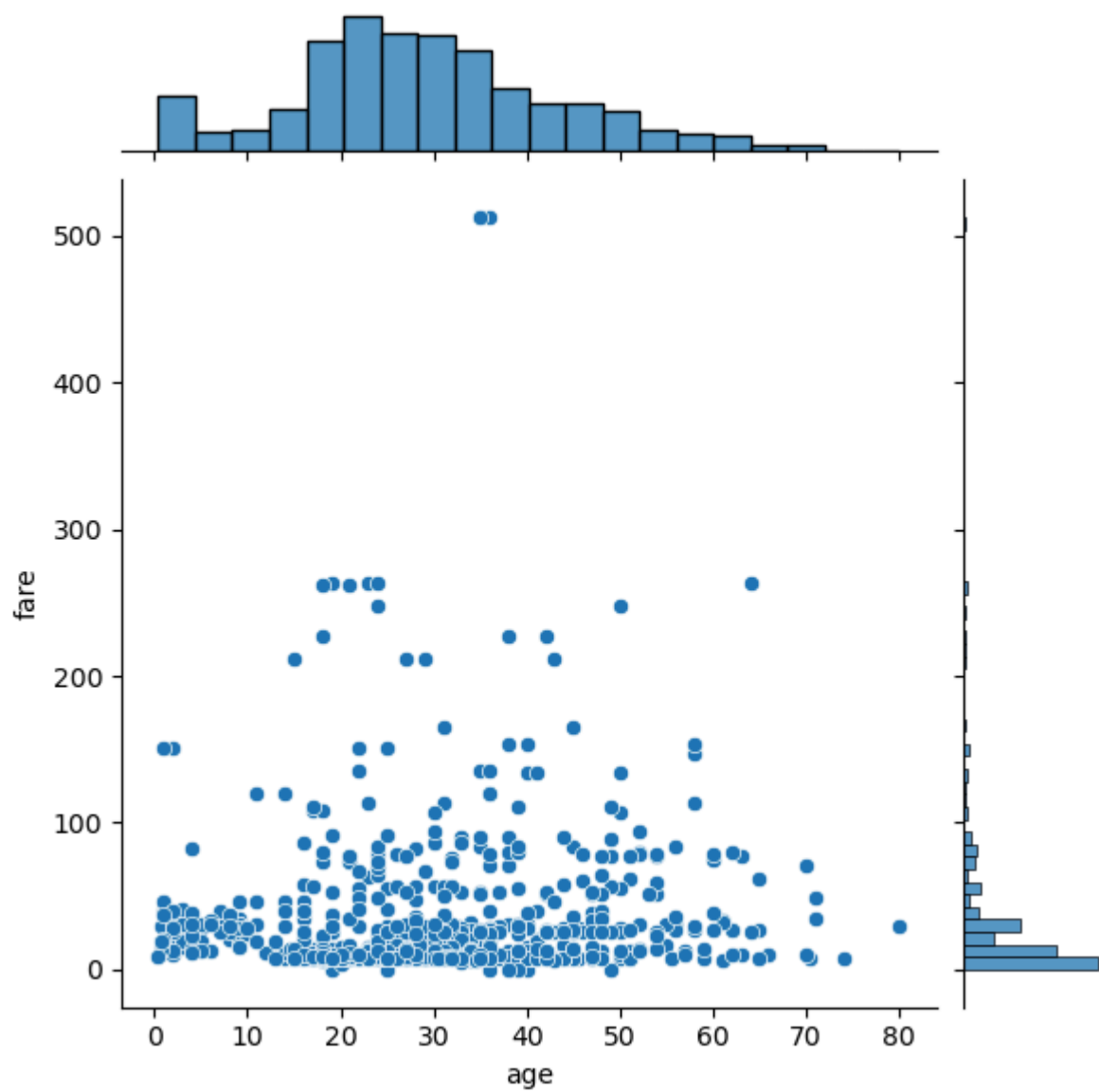
Out[7]:

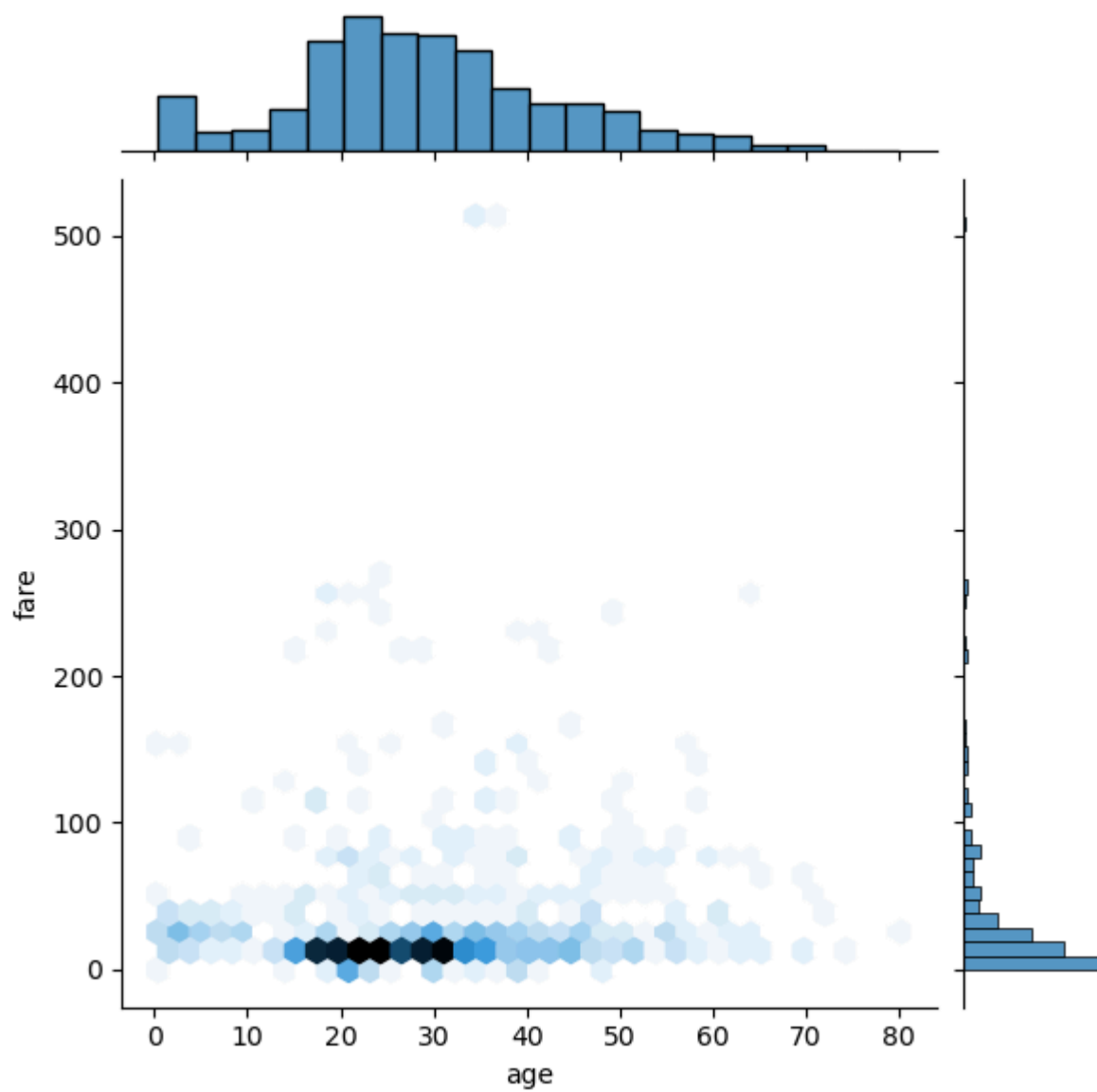
<seaborn.axisgrid.FacetGrid at 0x26a2f6a2490>



```
In [10]: sns.jointplot(x = titanic['age'], y = titanic['fare'], kind = 'scatter')
sns.jointplot(x = titanic['age'], y = titanic['fare'], kind = 'hex')
```

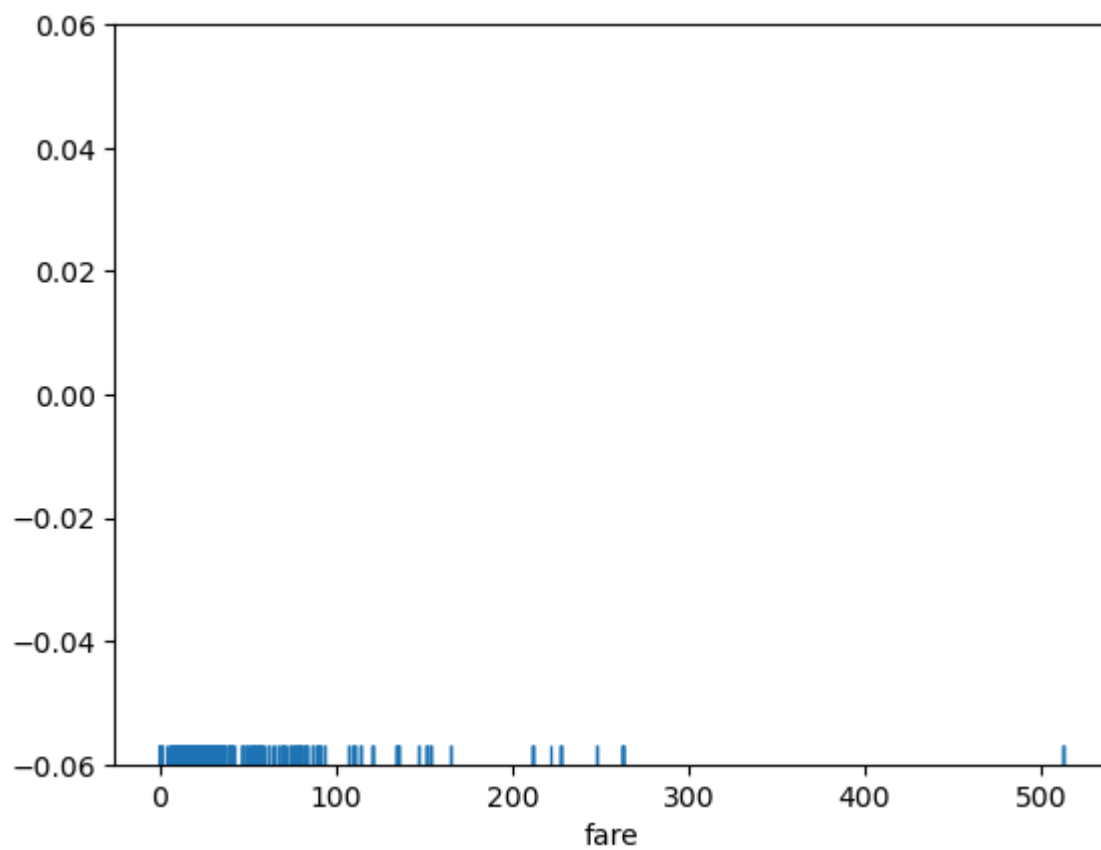
```
Out[10]: <seaborn.axisgrid.JointGrid at 0x26a2f7a5480>
```





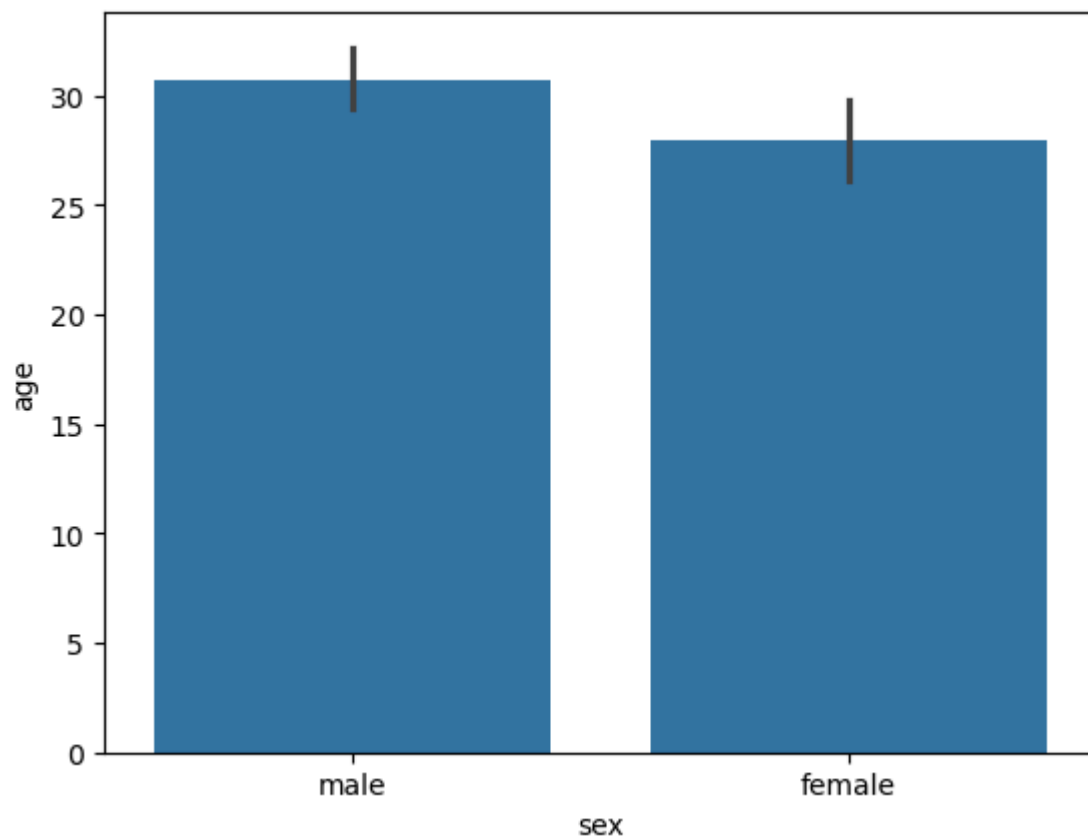
```
In [11]: sns.rugplot(titanic['fare'])
```

```
Out[11]: <Axes: xlabel='fare'>
```



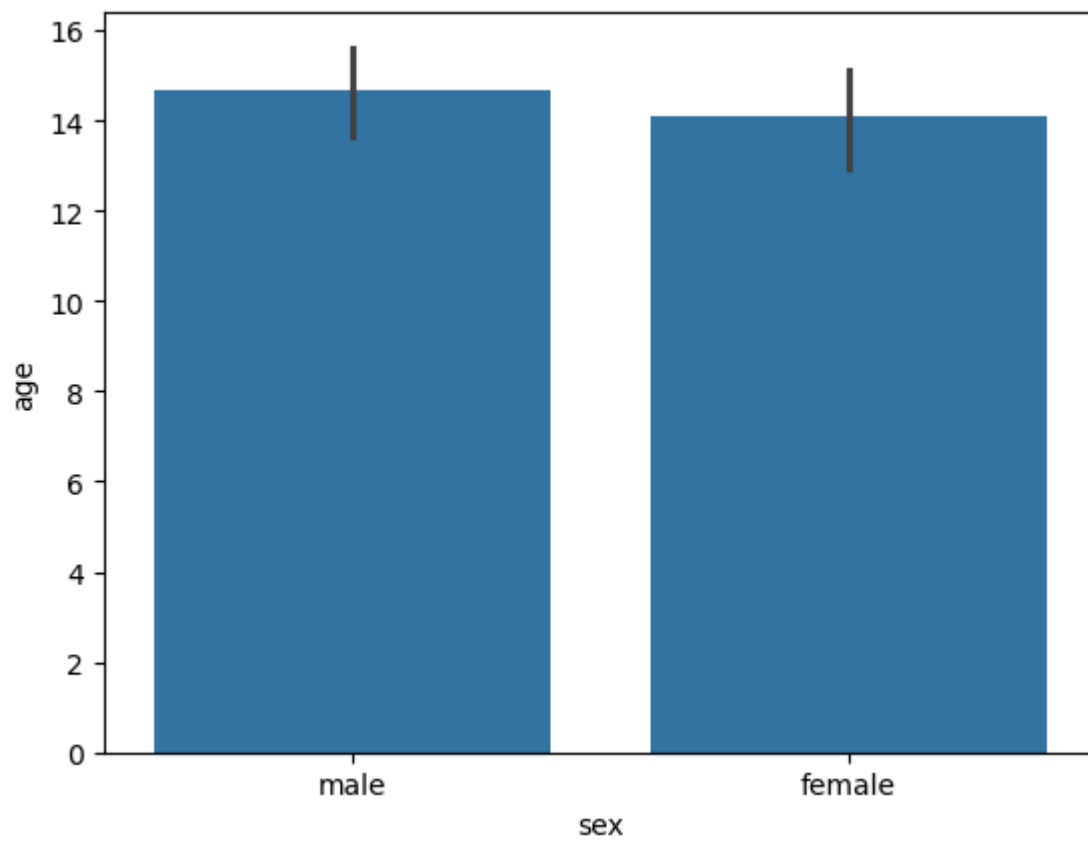
```
In [12]: sns.barplot(x = 'sex', y = 'age', data = titanic)
```

```
Out[12]: <Axes: xlabel='sex', ylabel='age'>
```



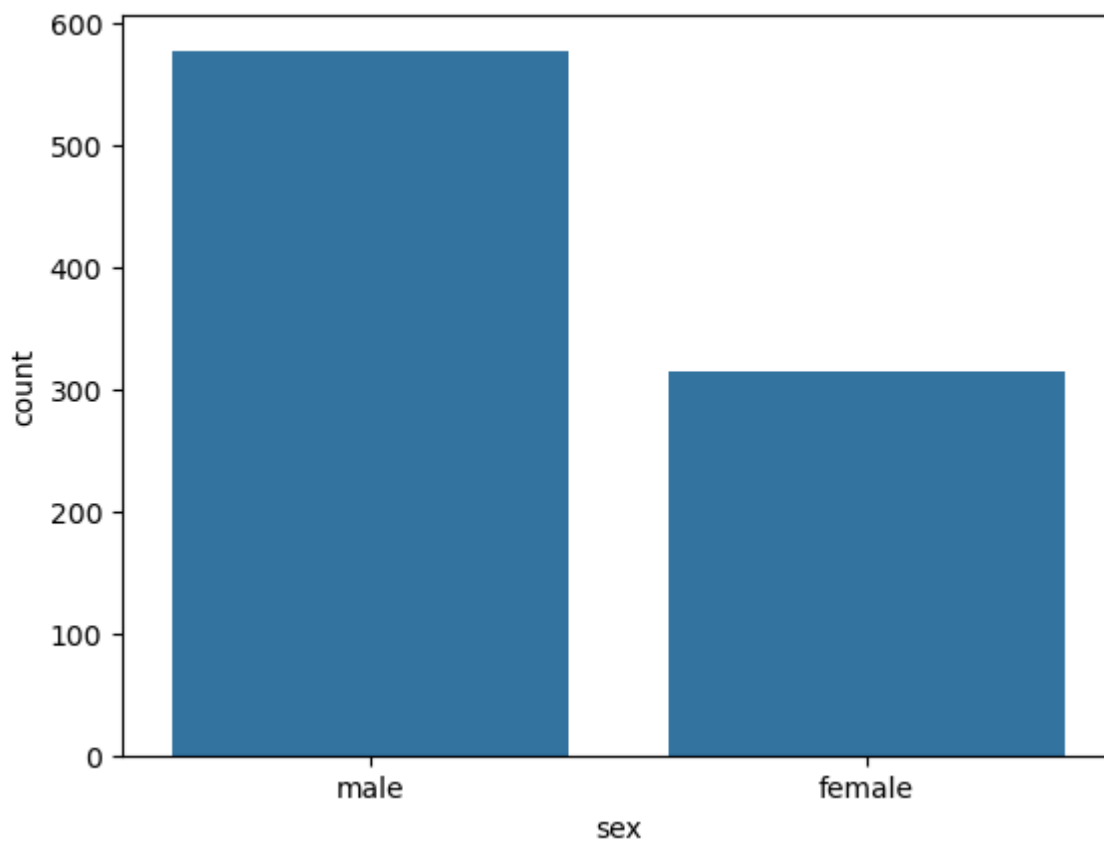
```
In [13]: sns.barplot(x = 'sex', y = 'age', data = titanic, estimator = np.std)
```

```
Out[13]: <Axes: xlabel='sex', ylabel='age'>
```



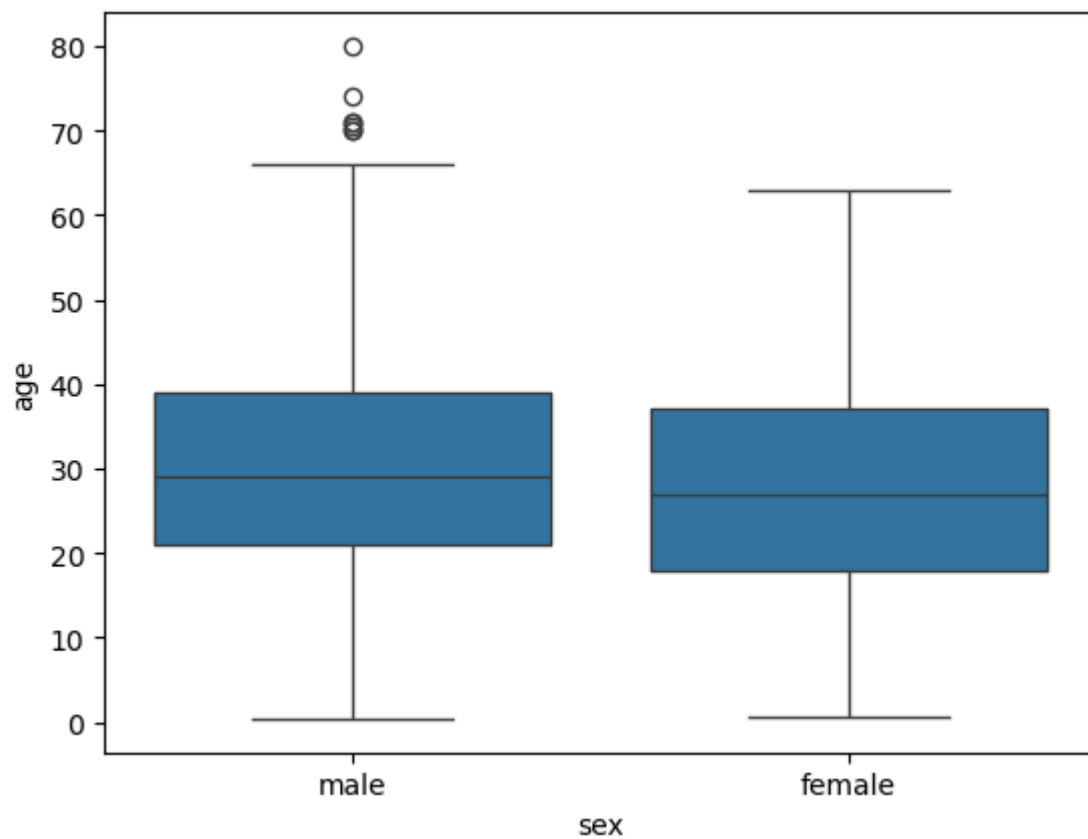
```
In [14]: sns.countplot(x = 'sex', data = titanic)
```

```
Out[14]: <Axes: xlabel='sex', ylabel='count'>
```



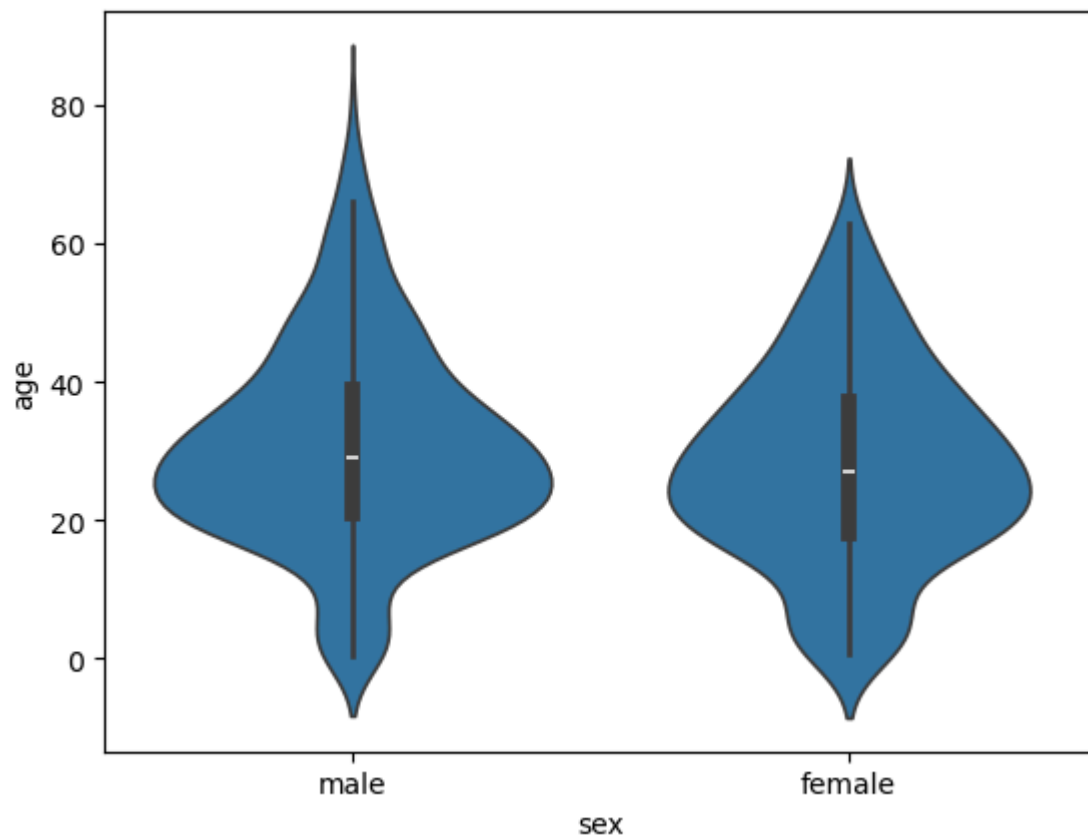
```
In [15]: sns.boxplot(x = 'sex', y = 'age', data = titanic)
```

```
Out[15]: <Axes: xlabel='sex', ylabel='age'>
```



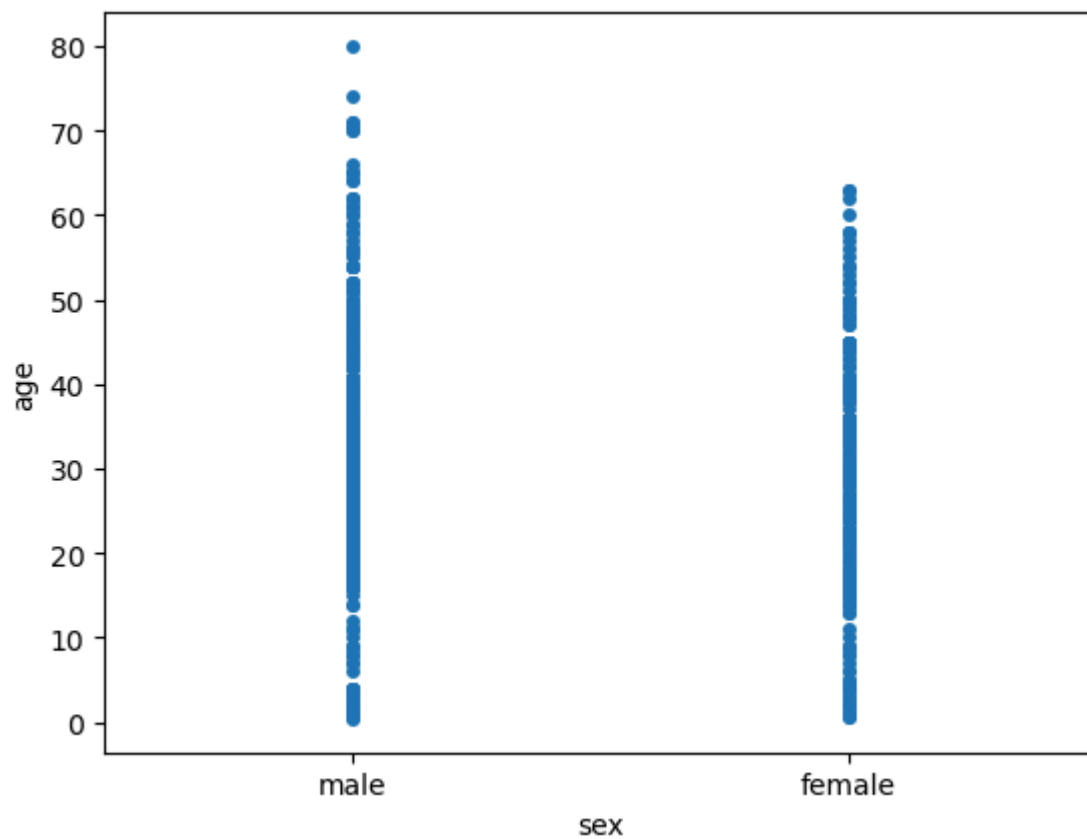
```
In [16]: sns.violinplot(x = 'sex', y = 'age', data = titanic)
```

```
Out[16]: <Axes: xlabel='sex', ylabel='age'>
```



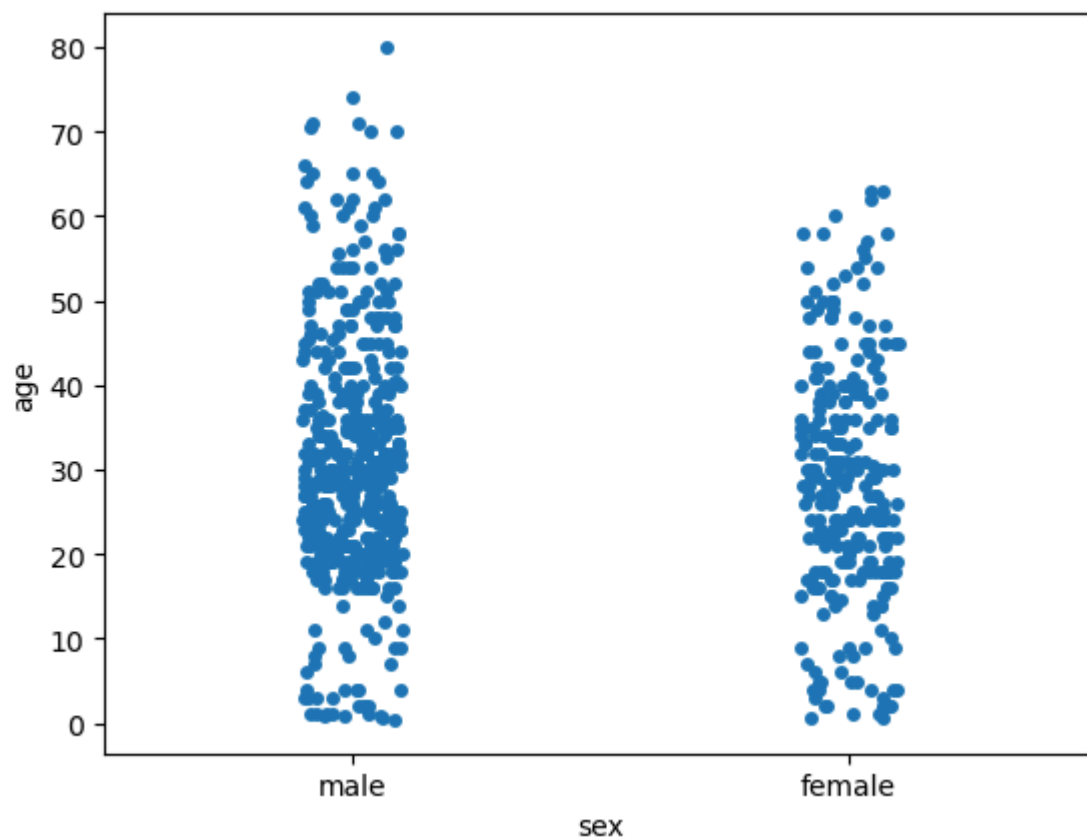
```
In [17]: sns.stripplot(x = 'sex', y = 'age', data = titanic, jitter=False)
```

```
Out[17]: <Axes: xlabel='sex', ylabel='age'>
```

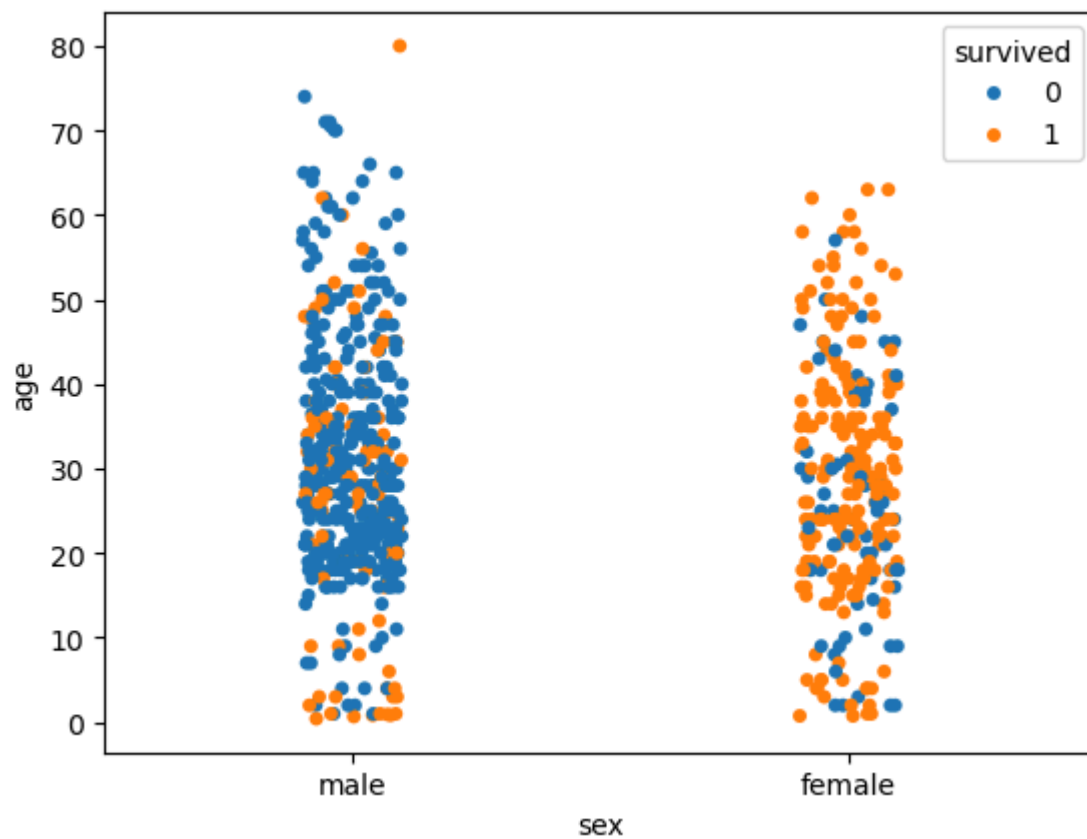
```
In [18]: sns.stripplot(x = 'sex', y = 'age', data = titanic, jitter=True)
```

```
Out[18]: <Axes: xlabel='sex', ylabel='age'>
```



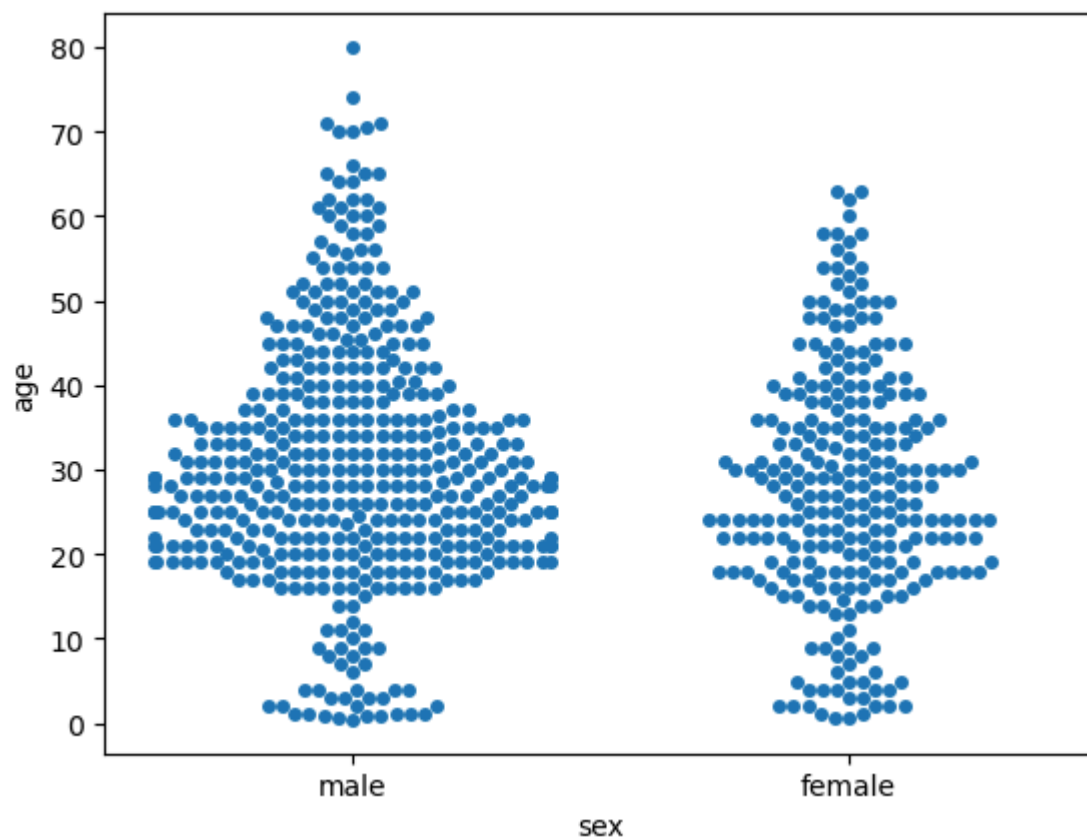
```
In [19]: sns.stripplot(x = 'sex', y = 'age', data = titanic, jitter=True, hue = 'survived')
```

```
Out[19]: <Axes: xlabel='sex', ylabel='age'>
```



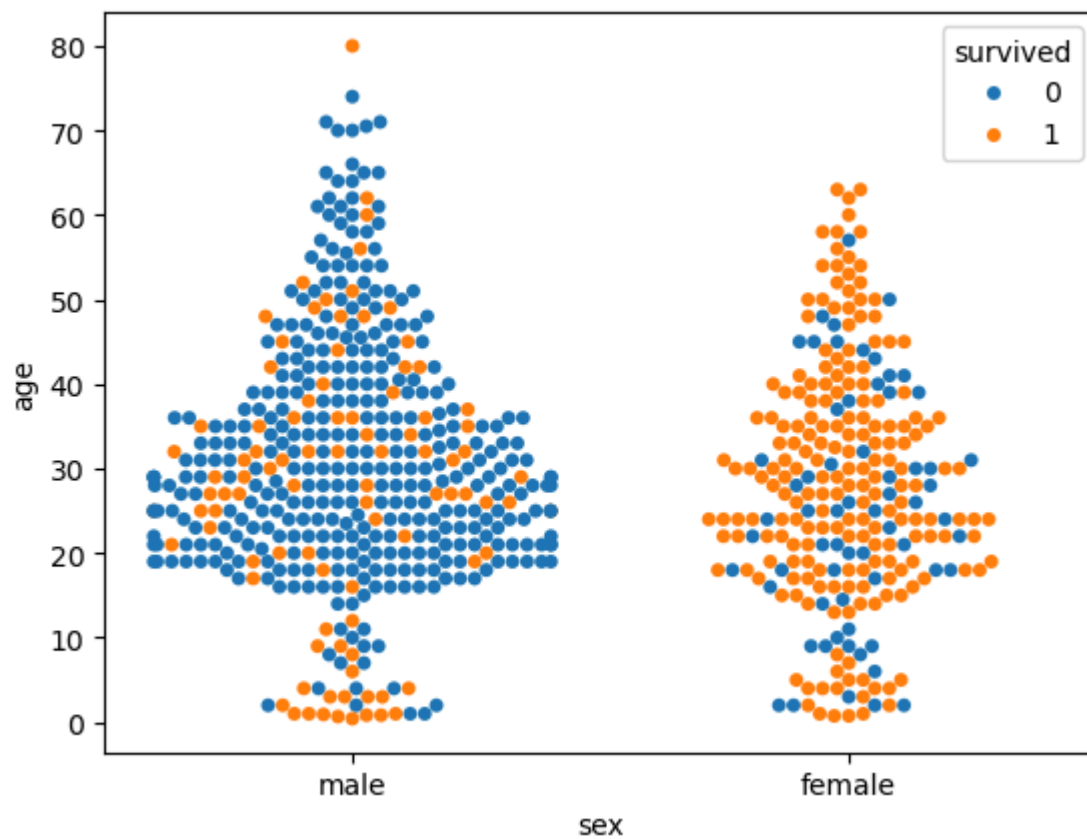
```
In [20]: sns.swarmplot(x = 'sex', y = 'age', data = titanic)
```

```
Out[20]: <Axes: xlabel='sex', ylabel='age'>
```



```
In [21]: sns.swarmplot(x = 'sex', y = 'age', data = titanic, hue = 'survived')
```

```
Out[21]: <Axes: xlabel='sex', ylabel='age'>
```



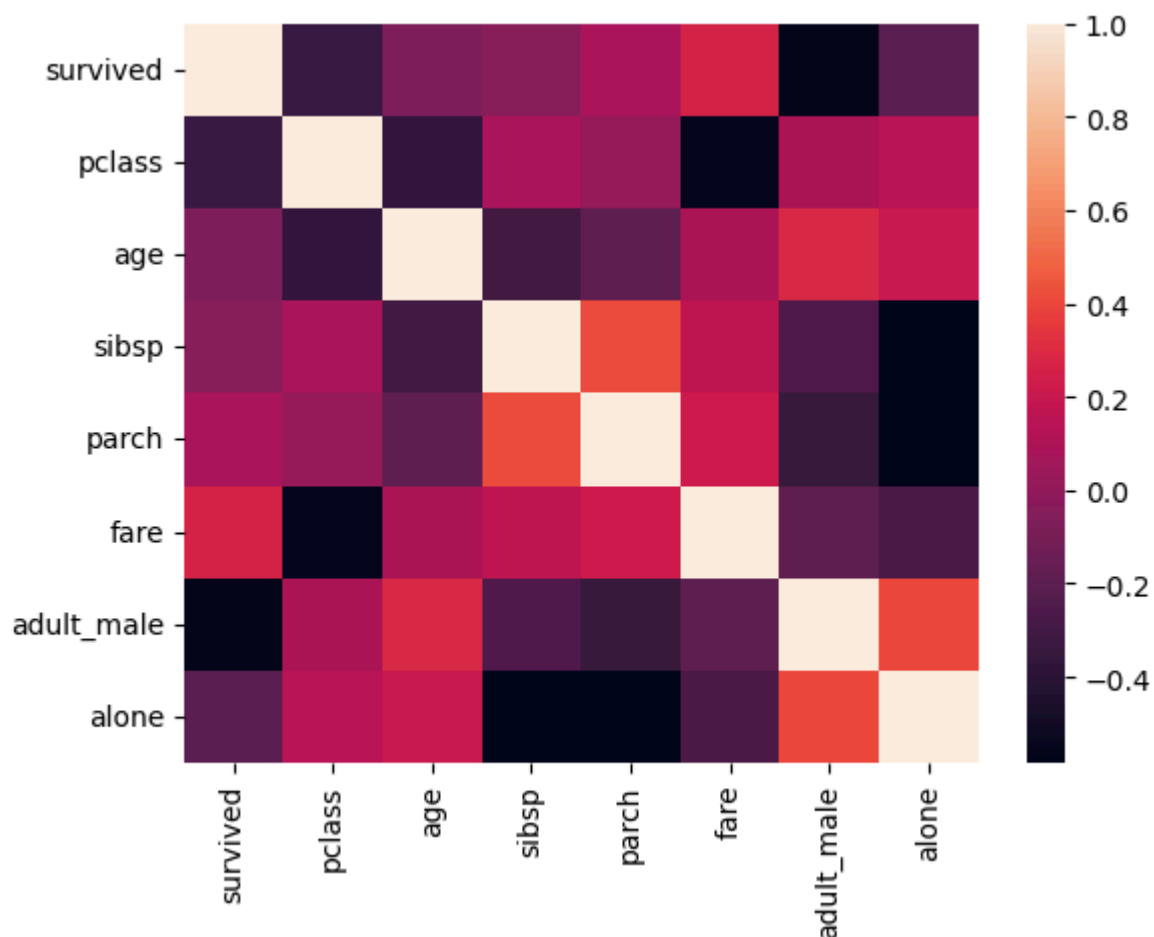
```
In [22]: titanic.corr(numeric_only= True)
```

```
Out[22]:
```

	survived	pclass	age	sibsp	parch	fare	adult_male	alone
survived	1.000000	-0.338481	-0.077221	-0.035322	0.081629	0.257307	-0.557080	-0.203367
pclass	-0.338481	1.000000	-0.369226	0.083081	0.018443	-0.549500	0.094035	0.135207
age	-0.077221	-0.369226	1.000000	-0.308247	-0.189119	0.096067	0.280328	0.198270
sibsp	-0.035322	0.083081	-0.308247	1.000000	0.414838	0.159651	-0.253586	-0.584471
parch	0.081629	0.018443	-0.189119	0.414838	1.000000	0.216225	-0.349943	-0.583398
fare	0.257307	-0.549500	0.096067	0.159651	0.216225	1.000000	-0.182024	-0.271832
adult_male	-0.557080	0.094035	0.280328	-0.253586	-0.349943	-0.182024	1.000000	0.404744
alone	-0.203367	0.135207	0.198270	-0.584471	-0.583398	-0.271832	0.404744	1.000000

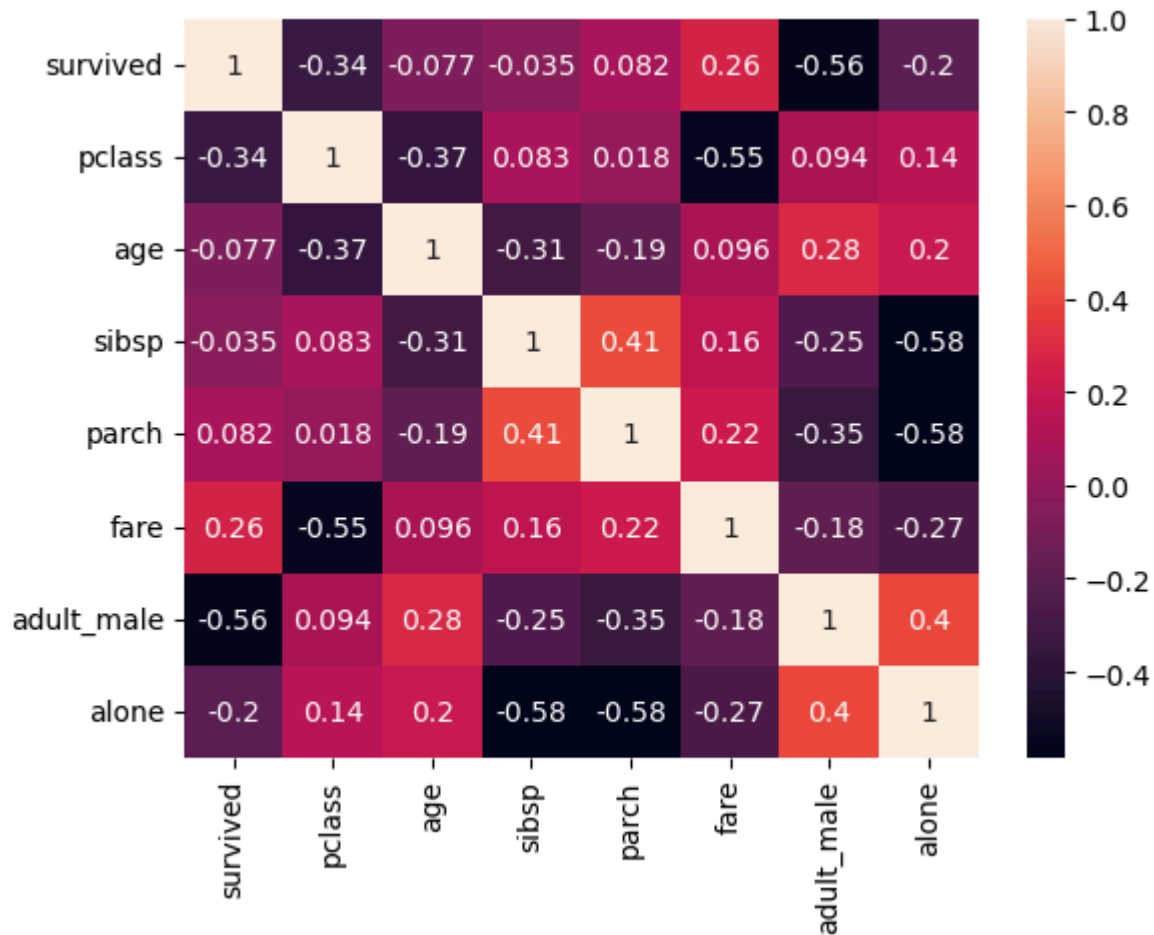
```
In [23]: sns.heatmap(titanic.corr(numeric_only= True))
```

```
Out[23]: <Axes: >
```



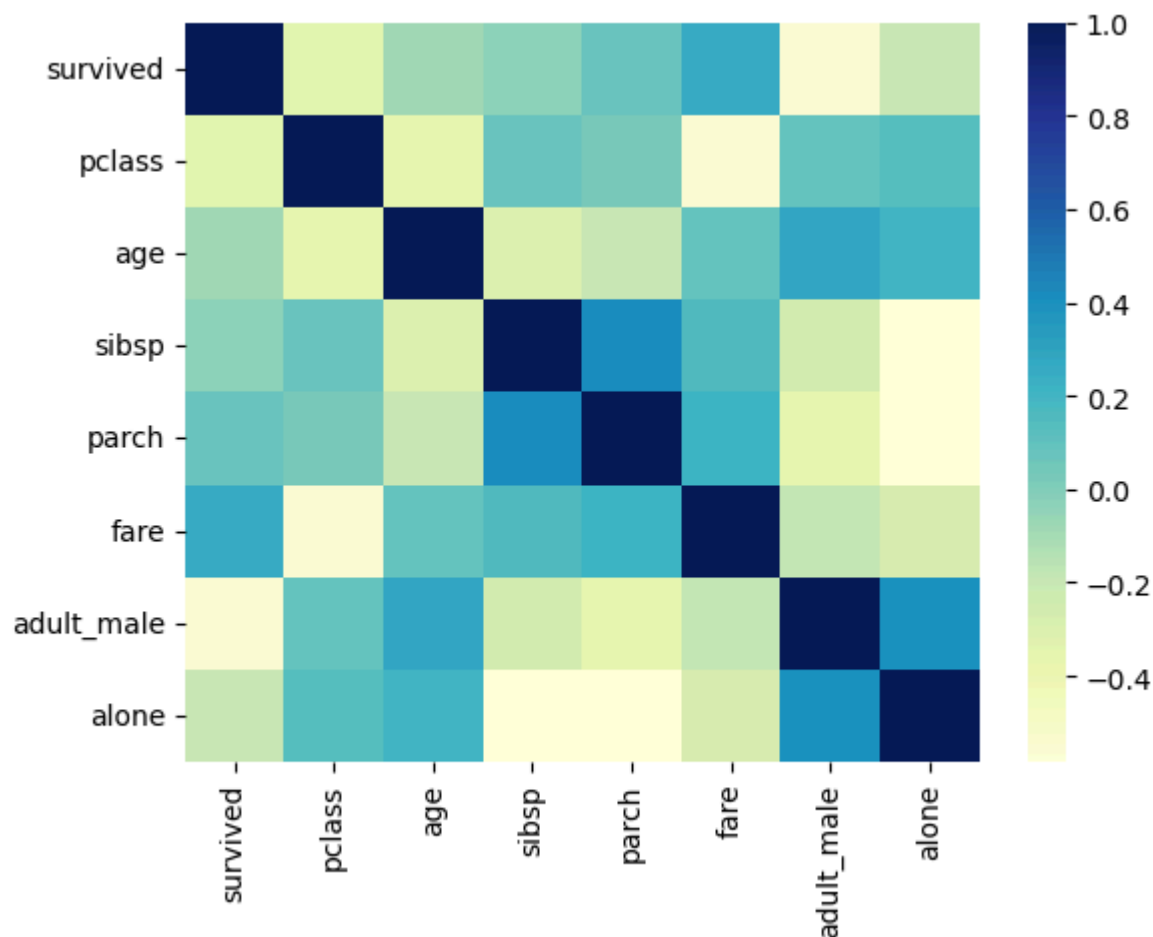
```
In [24]: sns.heatmap(titanic.corr(numeric_only= True), annot = True)
```

Out[24]: <Axes: >



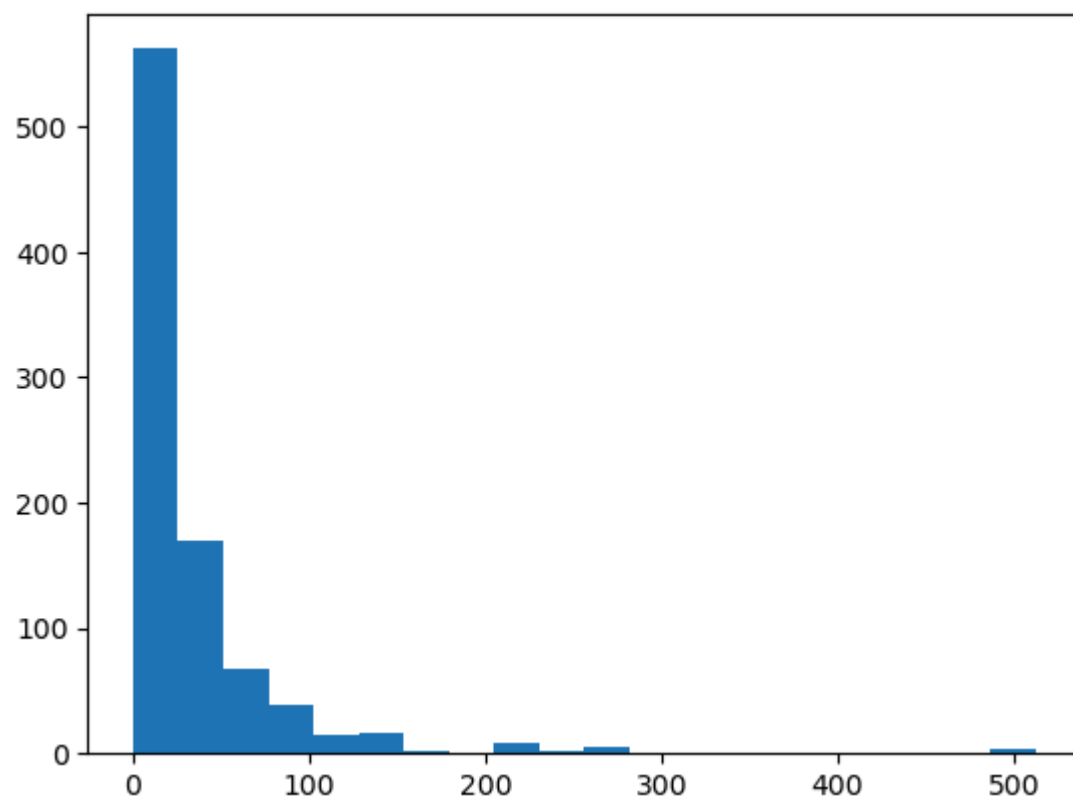
```
In [25]: sns.heatmap(titanic.corr(numeric_only= True), cmap = 'YlGnBu')
```

Out[25]: <Axes: >



```
In [26]: plt.hist(x = titanic['fare'], bins = 20)
```

```
Out[26]: (array([562., 170., 67., 39., 15., 16., 2., 0., 9., 2., 6.,
        0., 0., 0., 0., 0., 0., 0., 0., 3.]),
array([ 0., 25.61646, 51.23292, 76.84938, 102.46584, 128.0823 ,
        153.69876, 179.31522, 204.93168, 230.54814, 256.1646 , 281.78106,
        307.39752, 333.01398, 358.63044, 384.2469 , 409.86336, 435.47982,
        461.09628, 486.71274, 512.3292 ]),
<BarContainer object of 20 artists>)
```



In []: